ON DUTY TRACKING SYSTEM

DESIGN THINKING PROJECT

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ABSTRACT

The **OD Tracking Application** is a web-based system developed to simplify and digitize the On-Duty (OD) request, approval, and certificate verification process in educational institutions. The conventional manual process involves physical document submission, delays in obtaining approvals, difficulties in tracking request status, and challenges in maintaining student participation records. These limitations increase paperwork and create additional work for students and faculty members.

The proposed system provides a centralized digital platform that enables students to submit OD requests along with the necessary supporting documents and monitor their application status. Faculty members and Heads of Department (HODs) can review submitted requests, approve or reject applications, and provide remarks when necessary. Email notifications are integrated into the workflow to inform students about approval decisions and important status updates. Once students attend the respective events, they can upload participation certificates or other supporting proof through the application. Faculty members can then verify the uploaded certificates and update the verification status.

The application maintains organized digital records of OD requests, approval decisions, uploaded certificates, and verification details, making information easier to access and manage. By replacing paper-based procedures with a structured digital workflow, the system aims to improve transparency, accountability, and communication between students and faculty members. It also reduces manual follow-ups, minimizes the risk of misplaced documents, and simplifies the monitoring of pending requests and certificate submissions.

Overall, the OD Tracking Application offers an efficient and user-friendly solution for managing the complete On-Duty process, from request submission to final certificate verification. The system supports better record management and helps educational institutions improve the efficiency of their departmental administrative activities.

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
API	Application Programming Interface
CRUD	Create, Read, Update, Delete
CSS	Cascading Style Sheet
DB	DataBase
DBMS	Database Management System
DFD	Data Flow Diagram
FR	Functional Requirement
GUI	Graphical User Interface
HOD	Head of the Department
HTML	Hyper Text Markup Language
HTTP	Hyper Text Transfer Protocol
HTTPS	Hyper Text Transfer Protocol Secure
IDE	Integrated Development Environment
JSON	Java Script Object Notation
OD	On Duty
PO	Program Outcome
PSO	Program Specific Outcome
RBAC	Role Based Access Control
SMTP	Simple Mail Transfer Protocol
UI	User Interface
UML	Unified Modeling Language
URL	Uniform Resource Locator
UX	User Experience
XLSX	Excel Open XML Spreadsheet

CHAPTER 1

INTRODUCTION

1.1 Background Of The Project

In educational institutions, students participate in various academic and extracurricular activities such as workshops, seminars, technical symposiums, conferences, sports events, and industrial visits. Participation in these activities may require students to obtain On-Duty (OD) permission to attend events during regular academic hours. Traditionally, the OD process involves submitting applications, obtaining approvals from faculty members or department authorities, and maintaining records manually. This approach can be time-consuming and may lead to delays in approval, difficulties in tracking application status, and challenges in managing supporting documents. As the number of students and requests increases, maintaining accurate and accessible records becomes more difficult.

To overcome these challenges, the OD Tracking Application is developed as a web-based platform to digitize and organize the OD request management process. The system enables students to submit OD requests and relevant documents, while authorized faculty members can review applications, update their status, and manage approval-related activities. It also supports the submission and verification of participation certificates, helping maintain organized records of completed events. By providing a centralized platform for request management, the application aims to reduce paperwork, improve communication, enhance transparency, and minimize administrative workload. The project demonstrates how web technologies can be used to improve the efficiency and reliability of administrative processes in educational institutions.

1.2 Problem Statement

The traditional On-Duty (OD) request management process in educational institutions often involves manual application submission, physical approvals, and separate record maintenance. These procedures can be time-consuming and may result in delays, misplaced documents, and difficulties in tracking the status of applications. Students may need to contact faculty members repeatedly to know whether their requests have been approved, while faculty members and department authorities may face challenges in managing multiple requests and maintaining accurate records. Additionally, handling participation certificates and verifying supporting documents manually can increase the administrative workload.

Therefore, there is a need for a centralized web-based system that simplifies the OD request process and improves coordination between students, faculty members, and administrators. The proposed OD Tracking Application aims to address these issues by providing digital request submission, approval management, status tracking, and certificate verification. By organizing OD-related information in a common platform, the system seeks to reduce paperwork, improve transparency, minimize manual effort, and make the overall OD management process more efficient and reliable.

1.3 Objectives of the Project

The main objective of the OD Tracking Application is to develop a web-based platform that simplifies and streamlines the On-Duty request management process in educational institutions. The system aims to provide a centralized interface for students to submit OD requests and supporting documents, while enabling authorized faculty members and administrators to review applications and manage approval activities efficiently. It also aims to provide request status tracking, support participation certificate submission and verification, and maintain organized records for easy access and reference.

1.4 Scope of the Project

The scope of the OD Tracking Application is to develop a web-based platform that simplifies the management of On-Duty requests in educational institutions. The system enables students to submit OD applications along with the required supporting documents and allows authorized faculty members and department authorities to review requests and manage approval decisions. It also provides facilities for tracking request status, maintaining OD-related records, and handling participation certificates for verification. The application aims to provide a centralized and user-friendly interface that improves coordination between students, faculty members, and administrators.

The project focuses on reducing manual paperwork, improving the accessibility of information, and making the OD request process more organized and transparent. It supports digital record management and can provide email notifications where the feature is implemented. However, the system is primarily limited to OD request processing and related document management; it does not replace the complete academic administration system of the institution. Advanced features such as mobile application support, QR-based verification, automated reminders, advanced analytics, and integration with other institutional systems may be considered for future development.

1.4.1 Included Scope

The OD Tracking Application includes the following functionalities:

1. OD Request Submission: Allows students to submit OD applications with the required details and supporting documents
2. Faculty Approval: Enables authorized faculty members to review, approve, or reject OD requests.
3. Request Status Tracking: Allows students to monitor the progress and current status of their OD applications.

4. Document Management: Supports uploading and managing documents related to OD requests.
5. Certificate Verification: Enables students to upload participation certificates and authorized faculty members to verify them.
6. Email Notifications: Provides email notifications for relevant OD request updates where configured
7. Student Management: Supports the management of student information and accounts through the available administrative functions.
8. Faculty Management: Supports faculty account and information management according to the system's permissions.
9. Centralized Record Maintenance: Organizes OD request details, approval statuses, and related records for convenient access.
10. Role-Based Access: Provides access to application functions according to the roles and permissions assigned to students, faculty members, and administrators
11. Reduced Paperwork: Minimizes dependence on manual applications and physical record maintenance.
12. Improved Transparency: Provides a more organized and traceable process for managing OD requests and related activities.

1.4.2 Scope Limitations

The OD Tracking Application has the following limitations:

1. Internet Dependency: The application requires a stable internet connection to access its web-based functionalities.
2. User Input Accuracy: The accuracy of OD records depends on users providing correct information and valid supporting documents.
3. Approval Dependency: OD requests must be reviewed and approved by authorized faculty members or department authorities.
4. Certificate Authenticity: Uploaded participation certificates require manual verification by authorized personnel unless automated verification is implemented.

5. Notification Dependency: Email notifications depend on the proper configuration and availability of the email service.
6. Limited Integration: The application may not be integrated with other institutional systems, such as attendance management or student information systems.
7. Security and Access Restrictions: Access to student information and uploaded documents depends on the effectiveness of the implemented authentication and authorization mechanisms.
8. Storage Limitations: The availability and retention of uploaded documents depend on the configured file-storage system and its capacity.
9. Platform Limitation: The current system is primarily designed as a web application and may not provide a separate mobile application.
10. Future Enhancements: Features such as QR-based verification, automated reminders, advanced analytics, and comprehensive report generation may require further development.

1.5 Existing System

In the existing On-Duty (OD) management system, students generally submit OD applications manually or through basic digital communication methods to obtain permission for participating in academic and extracurricular activities. The applications are reviewed by faculty members and department authorities, and approval decisions are communicated through direct interaction, email, or other communication channels. Supporting documents and participation certificates may be maintained separately in physical files, folders, or spreadsheets. Students often need to contact the concerned faculty members to know the status of their requests, while faculty members must manually review applications and maintain records of approvals and student participation.

1.6 Limitations of Existing System

The traditional On-Duty (OD) request management system has several limitations, including time-consuming application submission and approval procedures, difficulty in tracking request status, and the possibility of losing or

misplacing important documents. Maintaining applications, approval details, and participation certificates manually can lead to errors, duplicate records, and difficulties in retrieving information. Students may need to contact faculty members repeatedly to obtain updates, while faculty members and department authorities must spend additional time reviewing applications, verifying certificates, and maintaining records. Furthermore, the lack of a centralized tracking mechanism can reduce transparency and cause communication delays. These challenges highlight the need for a web-based OD Tracking Application that simplifies request processing, improves document management, reduces administrative workload, and provides a more organized and transparent workflow

Table 1.1 Limitations of existing systems

Existing	Primary capability
Time-consuming record maintenance	Centralized record management
Delayed communication	Email notifications
Manual OD application submission	Online OD request submission
Scattered document storage	Organized document management

1.7 Proposed System

The proposed OD Tracking Application is a web-based platform developed to simplify and digitize the On-Duty request management process in educational institutions. It allows students to submit OD requests along with the required supporting documents. Authorized faculty members and department authorities can review applications, approve or reject requests, and update their status through the application. This centralized approach helps students track their requests and reduces the need for repeated manual enquiries.

The system also supports the digital submission and verification of participation certificates, helping maintain organized records of OD-related activities. Email notifications, where configured, help communicate important updates to users. By bringing request submission, approval management, status tracking, and document handling into a common platform, the proposed system aims to reduce paperwork,

improve transparency, and minimize administrative workload. Overall, it provides a more structured and convenient approach to managing OD requests while allowing scope for future enhancements.

1.8 Advantages of Proposed System

The OD Tracking Application offers several advantages over traditional OD request management methods by providing a centralized and organized digital platform. It reduces paperwork by allowing students to submit OD requests and supporting documents online, while authorized faculty members can review applications and manage approval decisions more efficiently. The system improves transparency by enabling students to track request statuses and helps maintain organized records of applications and participation certificates. Email notifications, where configured, improve communication between students and faculty members, while digital document handling simplifies certificate submission and verification. Furthermore, the application can reduce repetitive administrative tasks, minimize errors associated with manual record maintenance, and improve the accessibility of OD-related information for authorized users. Overall, the proposed system aims to make the OD management process more convenient, reliable, and efficient while providing a foundation for future enhancements and integration with other institutional systems.

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction

A literature review helps in understanding existing approaches, technologies, and research related to the development of web-based management systems. In educational institutions, digital platforms are increasingly used to manage administrative activities, student information, document processing, and approval workflows. Reviewing these approaches helps identify the benefits of automation, centralized data management, role-based access, and electronic communication. For the OD Tracking Application, the literature review focuses on web-based request management, workflow automation, document handling, notification mechanisms, and digital record maintenance. These concepts provide a foundation for designing a system that enables students to submit On-Duty requests, allows authorized faculty members to review and process applications, and supports request tracking and participation certificate verification. The review also helps identify limitations in traditional manual procedures and establishes the need for a centralized application that improves accessibility, transparency, and administrative efficiency.

2.2 Review of the related work

[1] Pinho, Franco and Mendes (2018)

Pinho, Franco, and Mendes [1] examined the role of web portals in improving information management within higher education institutions. The researchers reviewed existing studies and identified key areas, including the software used in web portals, their organizational benefits, technology acceptance, and information management and storage. The study highlighted that web portals can improve communication, provide access to relevant and updated information, and support decision-making in educational institutions. These findings are relevant to the

proposed OD Tracking Application, which uses a centralized web-based platform to manage on-duty requests, approval processes, request status, and certificate records. However, the study primarily focused on web portals and information management in higher education rather than specifically on OD request approval and certificate verification. Therefore, the proposed application addresses this specific administrative requirement by integrating request submission, approval, notification, and certificate verification into a single workflow.

[2] Mardaniyan (2019)

Mardaniyan [2] presented a web-based system for managing student information and academic records in educational institutions. The study focused on centralizing student data through a structured database and a role-based web interface, enabling authorized users to manage student profiles, course details, attendance, marks, and reports. The system aimed to reduce the problems associated with maintaining information in separate spreadsheets and paper-based records, such as data duplication, inconsistent updates, and delays in retrieving information. The study also emphasized authentication, data integrity, controlled access, and maintainability as important aspects of an effective information management system. These concepts are relevant to the proposed OD Tracking Application, which centralizes on-duty requests, approval decisions, status updates, and participation certificates in a single platform. However, the study primarily addresses general student information management rather than the specific workflow of OD request approval and certificate verification. The proposed application extends these general principles to support the complete OD management process.

[3] Falebita (2022)

Falebita [3] proposed a secure web-based platform for managing student and staff information at the Nigeria Police Academy. The study identified several limitations of paper-based record management, including difficulties in retrieving and updating records, data inconsistency, and duplication of information. To address these issues, the researcher developed a web-based system using the Laravel 5.5 PHP framework, with the objective of improving information accessibility and

protecting the application against common web security vulnerabilities. The system supported the management of student information, staff details, course registration, course materials, and academic results. This study is relevant to the proposed OD Tracking Application because both systems aim to reduce manual administrative work, centralize institutional records, and improve the accessibility and reliability of information. However, Falebita's system primarily focuses on student information management and web application security, whereas the OD Tracking Application focuses on the complete on-duty workflow, including request submission, faculty and HOD approval, email notifications, and participation certificate verification. Thus, the proposed application applies similar principles of digital information management to a specific administrative process in educational institutions.

[4] Destini and Tony (2024)

Destini and Tony [4] developed a web-based Document Management System to improve document administration in higher education institutions. The study addressed the problems associated with manual document handling, including delays, inefficiencies, and a lack of transparency in approval procedures. The proposed system incorporated Hierarchical Role-Based Access Control (HRBAC), which assigns access permissions according to the responsibilities and positions of users within an organizational hierarchy. Developed using the Laravel framework, PHP, HTML, SCSS, JavaScript, and MySQL, the system supported document submission, approval tracking, and administrative workflow management. The researchers evaluated usability using the System Usability Scale (SUS) and reported a score of 78.33, indicating a good level of usability. This study is relevant to the proposed OD Tracking Application because both systems aim to digitize document management, establish structured approval workflows, and improve transparency in institutional administration. However, the study focuses on document administration for student organizations, whereas the OD Tracking Application specifically manages student on-duty requests, faculty and HOD approvals, email notifications, and participation certificate verification. Therefore, the proposed application adapts

similar workflow management principles to the specific requirements of OD administration.

[5] Adegoke and Olalekan (2026)

Adegoke and Olalekan [5] proposed a web-based system to automate student clearance procedures across multiple departments in higher education institutions. The system was developed using the Django framework and MySQL database, incorporating role-based access control, document submission, automated email notifications, real-time clearance status tracking, and QR-code-based certificate verification. The study addressed the limitations of traditional clearance processes, such as delays, misplaced documents, inconsistent records, and poor coordination between departments. By integrating the activities of students, departmental officers, and administrators into a centralized platform, the system aimed to improve administrative efficiency, transparency, accountability, and document security. This study is closely related to the proposed OD Tracking Application because both systems provide digital document submission, structured approval workflows, status tracking, and email notifications. However, the clearance system focuses on completing institutional requirements across multiple departments, whereas the OD Tracking Application specifically manages on-duty requests, faculty and HOD approval decisions, and verification of participation certificates after events. Thus, the proposed application applies similar workflow automation principles to the specific requirements of on-duty management in educational institutions.

2.3 Comparative Analysis

Table 2.1 Literature survey comparison

Ref.	Author(s) / Year	Study focus	Methodology / approach	Reported inference
[1]	Pinho et al. (2018)	Web portals for information management in higher education	Systematic literature review of web portals in educational institutions	Web portals improve information accessibility, communication, and institutional information management.
[2]	Mardaniyan (2019)	Web-based student information management	Centralized student records managed through a web-based system	Digital information management simplifies record maintenance and retrieval.
[3]	Falebita (2022)	Secure web-based student information management	Web application for managing student and staff information with security considerations	Web-based systems help reduce paper-based record problems and improve information accessibility.
[4]	Destini and Tony (2024)	Document administration in student organizations	Web-based document management with hierarchical role-based access control	Role-based permissions and digital document workflows improve administration and document handling.
[5]	Adegoke and Olalekan (2026)	Online student clearance and multi-departmental workflow	Workflow-driven web system with document submission, approval tracking, and notifications	Centralized workflows can improve coordination, transparency, and progress tracking.

2.4 Comparative Analysis of the Proposed System

The proposed OD Tracking Application is designed to bring the major stages of OD management into a unified digital workflow. Unlike a process that depends entirely on physical applications or separate communication channels, the application supports online request submission, faculty review, status updates, and participation certificate handling. Its centralized approach can make OD-related information easier for authorized users to access and manage. The system also supports communication through configured notifications and provides a structured foundation for maintaining request records. However, the proposed application is specifically intended for OD management and should not be considered a complete replacement for all institutional administrative systems. Its benefits must be

evaluated according to the features implemented and the requirements of the institution.

Table 2.2 Comparative Analysis of OD Management Approaches

Feature	Higher Education Web Portals	Student Information System	Secure Student Information System	Document Management System	Student Clearance System	Proposed OD Tracking System
Centralized information management	No	Yes	Yes	Yes	Yes	Yes
Online document submission	General	Limited	Limited	Yes	Yes	Yes
Role-based access	General	Yes	Security-focused	Yes	Yes	Yes
Structured approval workflow	General	Limited	Limited	Yes	Yes	Yes
OD request submission	No	No	No	No	No	Yes
Faculty/HOD approval	No	No	No	Organization-based	Department-based	Yes
Request status tracking	General	Limited	Limited	Yes	Yes	Yes
Email notifications	Portal-dependant	Not established	Not established	Not established	Yes	Yes
Certificate upload and verification	No	No	No	General document handling	QR-based verification	Yes
Centralized OD record maintenance	General	Student records	Student records	Document records	Clearance records	Yes

2.5 Research Gap

Traditional OD management methods and disconnected digital approaches may not provide a single, organized workflow covering request submission, approval processing, status tracking, and participation certificate verification. When these activities are handled separately, students may have difficulty obtaining updates, while faculty members may spend additional time coordinating requests and maintaining records. The project addresses this practical need by developing a centralized web-based application that brings these activities together and provides role-appropriate access to information. The intended contribution of the OD Tracking Application is to improve the organization, accessibility, and traceability of OD-related activities within an educational institution. A stronger research gap should be finalized after reviewing and comparing specific published studies and existing systems.

CHAPTER 3

THEORETICAL BACKGROUND AND SYSTEM DESIGN

3.1 General

The theoretical background of the OD Tracking Application includes web-based application development, database management, user authentication, role-based access control, and workflow automation. The system design defines how students submit OD requests, how faculty members and HODs review and approve them, and how certificates are uploaded and verified after events. The application integrates the frontend, backend, database, and email notification modules to provide a centralized and organized platform for managing the complete OD process efficiently and transparently.

3.2 Web-based Application Development

Web-based application development involves creating applications that can be accessed through web browsers over a network. The OD Tracking Application follows a client-server architecture in which students, faculty members, and HODs interact with the system through a user interface, while the backend processes requests and manages data stored in the database. This approach enables centralized record management, easy access to OD requests, efficient approval workflows, and certificate verification without relying on physical documents.

3.3 Frontend Technologies

The frontend of the OD Tracking Application provides the user interface through which students, faculty members, and HODs interact with the system. HTML5 is used to structure the web pages, CSS3 is used to design the layout and appearance, and JavaScript is used to support interactive features and user actions. The interface enables students to submit OD requests, upload supporting documents, and track request status, while faculty members and HODs can review requests and verify

certificates. The frontend is designed to provide a simple, accessible, and user-friendly experience for all users.

3.4 Backend Development using Python and Flask

The backend of the OD Tracking Application is responsible for processing user requests, implementing application logic, managing approval workflows, and communicating with the database. Python provides the programming logic, while Flask serves as the web framework for handling HTTP requests, defining routes, and connecting frontend operations with backend functions. The backend manages OD request submission, approval and rejection, certificate uploads, status updates, and email notifications. It also validates user actions and ensures that operations are performed according to the permissions assigned to each user role. This backend structure supports reliable communication between the application interface and the database, enabling efficient and organized OD management.

3.5 Database Management using Azure SQL

Azure SQL is used as the database management system for the OD Tracking Application to store and manage structured information. It maintains student and faculty details, OD request records, approval statuses, and certificate-related information in an organized manner. The database allows the backend developed using Python and Flask to perform data insertion, retrieval, updating, and deletion operations as required. Its integration with the application supports centralized data management, reliable record access, and efficient processing of OD requests. Azure SQL also provides a cloud-based database solution that can support the application's deployment and future scalability.

3.6 User Authentication and Authorization

User authentication and authorization help protect the OD Tracking Application from unauthorized access. Authentication verifies a user's identity through login credentials, while authorization determines which functions the user can access based on their assigned role. Students can submit OD requests and upload certificates, whereas faculty members and HODs can review requests, manage

approvals, and verify submitted documents. These mechanisms help protect sensitive information, maintain controlled access to system functions, and ensure that users perform only the operations permitted by their roles.

3.7 Role-based Access Control

Role-Based Access Control (RBAC) is a security mechanism that assigns system permissions according to the responsibilities of each user. In the OD Tracking Application, students, faculty members, and Heads of Departments (HODs) are provided with access to specific functions based on their roles. Students can submit OD requests, upload certificates, and track their request status, while faculty members and HODs can review requests, make approval decisions, and verify certificates according to their assigned permissions. This approach helps prevent unauthorized operations, protects institutional information, and ensures the proper functioning of the OD management workflow.

3.8 OD Request Management

OD Request Management is a core module of the OD Tracking Application that enables students to submit On-Duty requests electronically. Students can provide the required event details and upload supporting documents for review. The submitted requests are stored in the database and made available to the authorized faculty members and HODs for approval or rejection. The module helps students track their request status and reduces paperwork, manual record maintenance, and delays in the traditional OD application process.

3.9 Approval Workflow Management

The Approval Workflow Management module handles the review and decision-making process for submitted OD requests. After a student submits a request, the authorized faculty member or HOD reviews the request details and supporting documents before approving or rejecting it. The system updates the request status based on the decision and notifies the student through email. This structured workflow improves transparency, reduces manual follow-ups, and helps ensure that OD requests are processed in an organized and timely manner.

3.10 File Upload and Document Storage

The File Upload and Document Storage module enables students to upload OD approval letters and supporting documents through the web application. It also supports the submission of participation certificates after the completion of an event for faculty verification. The system manages uploaded documents and maintains their association with the corresponding OD requests, allowing authorized users to access and review them when required. This module reduces dependence on physical documents, simplifies document retrieval, and improves the organization of OD-related records.

3.11 Certificate Verification

The Certificate Verification module enables faculty members and HODs to verify the participation certificates submitted by students after completing an event. The uploaded certificates are reviewed to confirm the student's participation and ensure that the required proof has been provided. Once verification is completed, the system records the verification status for future reference. This module reduces manual verification efforts, improves record accuracy, and helps maintain reliable documentation of student participation in on-duty activities.

3.12 Email Notification System

The Email Notification System keeps students and relevant faculty members informed about important updates in the OD approval process. When an OD request is approved or rejected, the system sends an email notification to the student, helping them know the decision without repeated manual follow-ups. This module improves communication between students and faculty members, reduces delays in conveying approval decisions, and makes the OD management process more transparent and efficient.

3.13 Student Management

The Student Management module maintains student information required for the OD Tracking Application. It enables authorized administrators or faculty members to manage student details and associate each student with the appropriate OD request records. Maintaining organized student information helps identify request owners, retrieve relevant records, and support accurate tracking of applications and certificates. This module improves the accessibility of student-related data and simplifies administrative activities within the institution.

3.14 Faculty and HOD Management

The Faculty and HOD Management module supports the participation of faculty members and Heads of Departments (HODs) in the OD approval process. It enables authorized faculty members and HODs to access submitted OD requests, review supporting documents, and approve or reject applications according to their responsibilities. The module also supports certificate verification and the maintenance of approval records. By assigning appropriate responsibilities to each role, it helps improve coordination, accountability, and efficiency in OD request management.

3.15 Request Status Tracking

The Request Status Tracking module enables students to monitor the progress of their OD applications through the web application. Each request displays its current status, such as pending, approved, or rejected, based on the decision made by the authorized faculty member or HOD. The status is updated when the approval decision is recorded, allowing students to understand the progress of their applications without repeatedly contacting faculty members. This module improves transparency, reduces manual follow-ups, and helps maintain an organized record of OD requests.

3.16 System Integration and Deployment

System integration combines the frontend, backend, database, and email notification modules to ensure that they function together as a complete application. The frontend communicates with the backend to submit OD requests and display their status, while the backend processes user actions and manages the associated records in the database. Email notifications support communication of approval decisions, and document management enables certificate submission and verification. Deployment makes the application accessible to authorized users through a web browser, allowing students, faculty members, and HODs to perform their respective activities through a centralized platform.

3.17 System Architecture

The system architecture of the OD Tracking Application describes the interaction between the major components of the application. It consists of a frontend interface, a backend application, a database, and an email notification module. Students interact with the frontend to submit OD requests, upload documents, and track their application status, while faculty members and HODs review requests and verify certificates. The backend processes user actions, applies access permissions, and communicates with the database to store and retrieve information. The email notification module communicates approval decisions to students. This architecture ensures coordinated communication between the components and supports efficient, centralized OD request management.

3.18 Data Flow Diagram

A Data Flow Diagram (DFD) represents the movement of information between users, system processes, and data storage components. In the OD Tracking Application, the DFD illustrates how students submit OD requests and supporting documents, how faculty members and HODs review applications, and how approval decisions are communicated to students. It also represents the storage of request details, the updating of request status, and the submission and verification of participation certificates. The DFD helps explain the system's functional structure and the flow of information between its different modules.

3.18.1 Level 0

The Level 0 Data Flow Diagram represents the OD Tracking Application as a single process interacting with external users. Students submit OD requests and supporting documents to the system and receive approval notifications and status updates. Faculty members and HODs provide approval decisions and verify participation certificates through the application. The system processes these inputs, maintains the relevant records in the database, and provides the required information to authorized users. This diagram presents a high-level overview of the interaction between users and the OD Tracking Application.

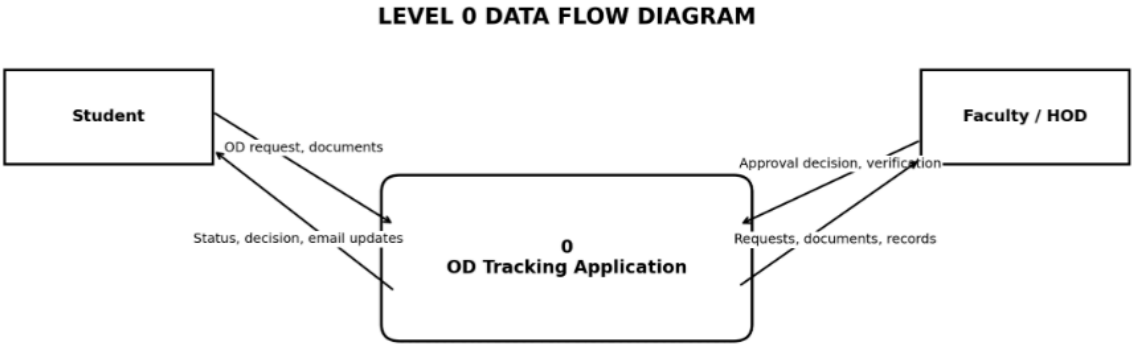


Figure 3.1 Data flow diagram – Level 0

3.18.2 Level 1

The Level 1 Data Flow Diagram illustrates the major processes of the OD Tracking Application, including OD request submission, request review and approval, certificate upload and verification, and status updates with email notifications. It shows how students provide request details and supporting documents, while faculty members and HODs review applications and verify certificates. The system stores request information, approval decisions, and certificate records in the database, enabling organized record maintenance and efficient tracking of the OD process.

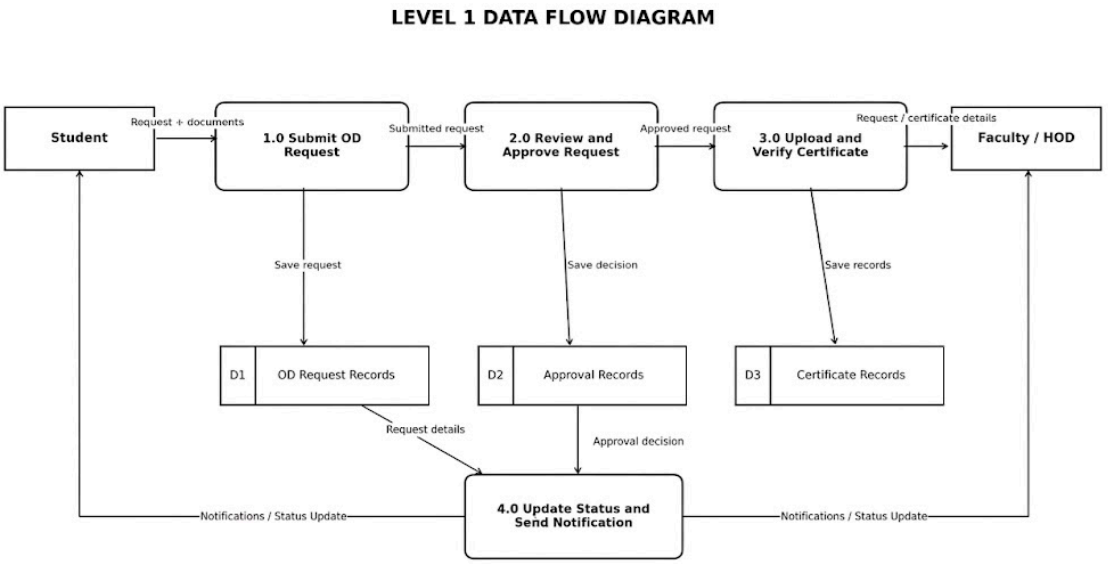


Figure 3.2 Data flow diagram – Level 1

3.18.3 Level 2

The Level 2 Data Flow Diagram provides a detailed representation of the internal processes of the OD Tracking Application. It breaks down OD request submission into entering request details, uploading supporting documents, validating information, and saving the request. It further illustrates the review and approval process, certificate uploading and verification, request status updates, and email notifications. These processes interact with the database to maintain request details, approval decisions, and certificate records, ensuring systematic and transparent OD management.

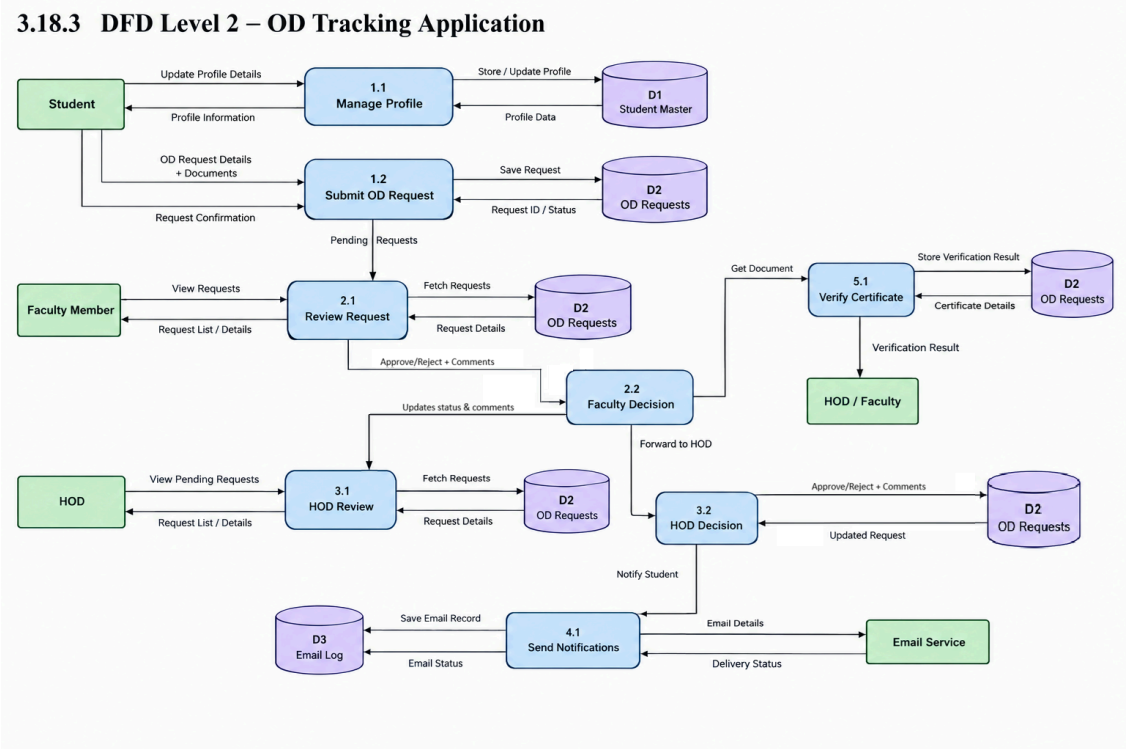


Figure 3.3 Data flow diagram – Level 2

3.19 Use Case Diagram

A Use Case Diagram represents the functional interactions between users and the system. In the OD Tracking Application, the main actors are students, faculty members, and Heads of Departments (HODs). Students can log in, submit OD requests, upload supporting documents, track request status, and submit participation certificates. Faculty members and HODs can review requests, approve or reject applications, and verify submitted certificates according to their assigned permissions. The system also sends email notifications regarding approval decisions and maintains relevant records. The use case diagram provides a clear overview of the system's functionality and the responsibilities of each user role.

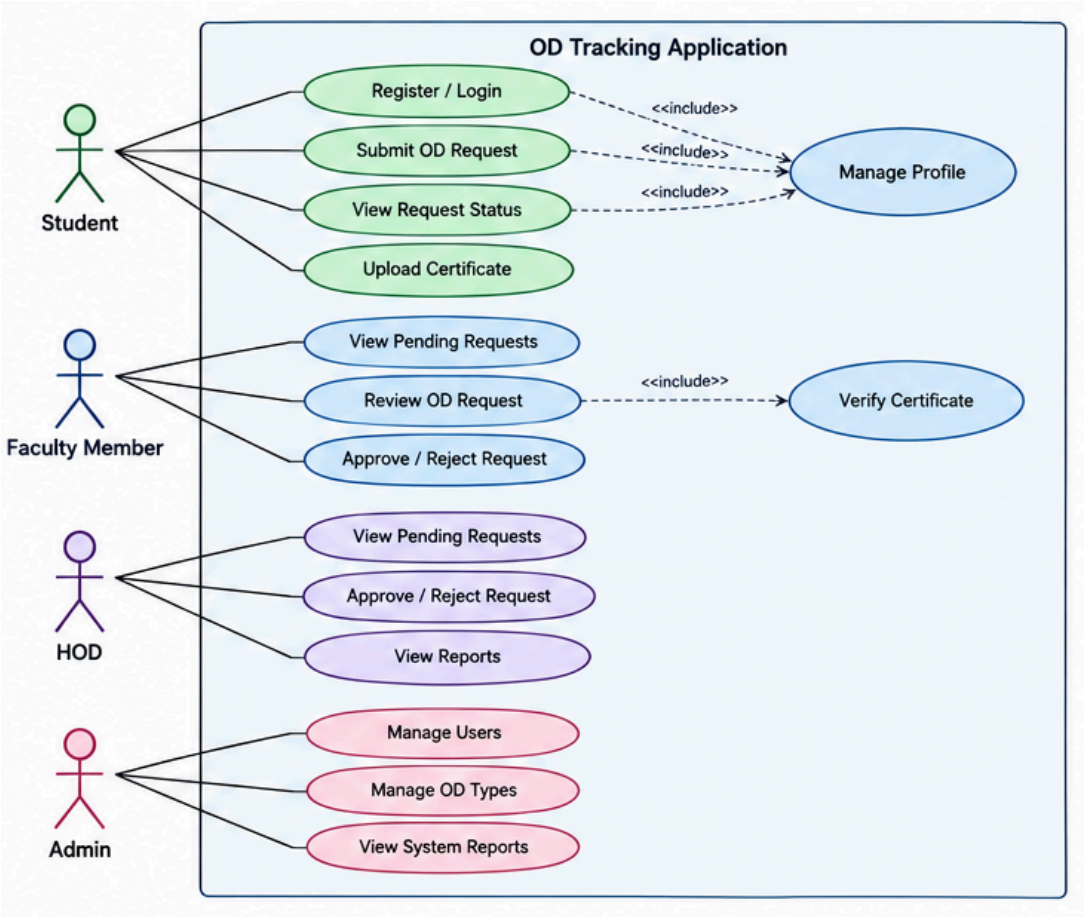


Figure 3.4 Use case diagram

3.20 Activity Diagram

The Activity Diagram represents the sequence of activities involved in the OD Tracking Application, from submitting an OD request to verifying the participation certificate. The process begins when a student logs in and submits an OD request with the required supporting documents. The faculty member or HOD reviews the request and either approves or rejects it. The system updates the request status and sends an email notification to the student. After participating in the event, the student uploads the certificate for verification. The faculty member or HOD verifies the submitted certificate, and the system updates the verification status and maintains the relevant records. This diagram illustrates the decision points and workflow involved in managing OD requests efficiently.

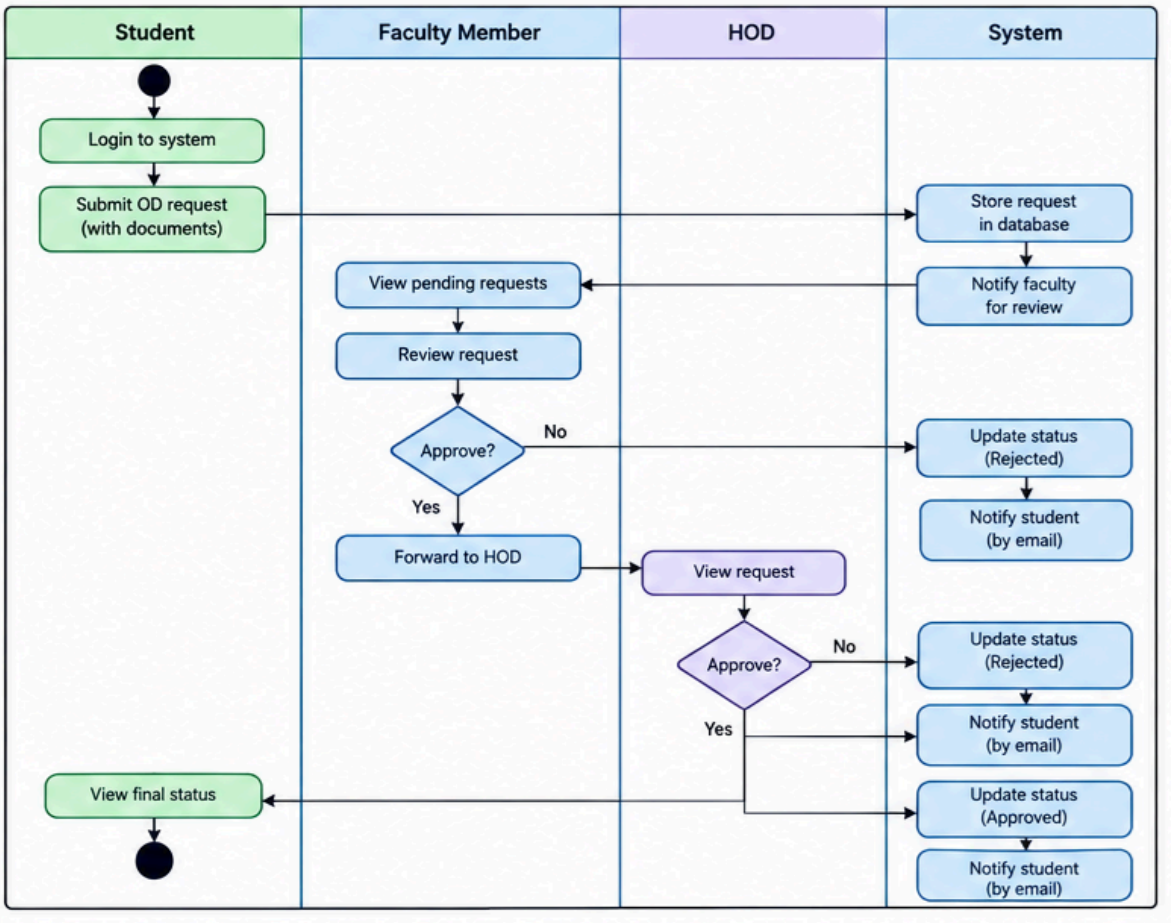


Figure 3.5 Activity diagram

3.21 Overall System Diagram

The Overall System Diagram illustrates the complete structure and interaction of the major components of the OD Tracking Application. It consists of the frontend interface, backend application, database, and email notification system. Students use the frontend to submit OD requests, upload supporting documents, track request status, and submit participation certificates. Faculty members and HODs review requests, make approval decisions, and verify certificates based on their assigned roles. The backend processes user actions, manages the workflow, and communicates with the database to store and retrieve information. The email notification system informs students about approval decisions and relevant updates. These integrated components provide a centralized platform for efficient and transparent OD management.

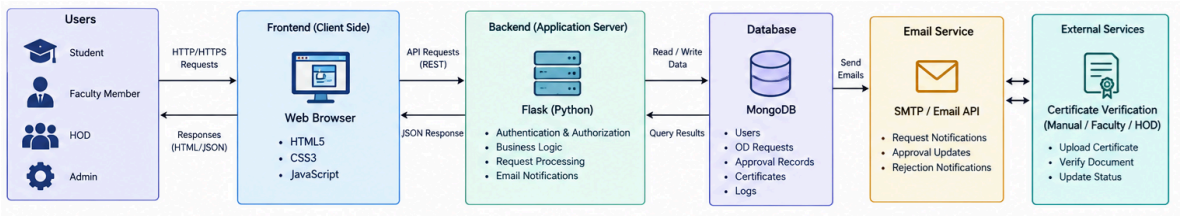


Figure 3.6 Overall system diagram

3.22 Core Conceptual Classes

The core conceptual classes of the OD Tracking Application represent the main entities involved in the system and their relationships. The User class manages login credentials and user roles, while the Student class maintains student-related information. The Faculty and HOD classes represent the users responsible for reviewing OD requests and making approval decisions. The ODRequest class stores request details, event information, and approval status, while the Document class manages uploaded OD letters and participation certificates. The CertificateVerification class records certificate verification details, and the Notification class represents email updates sent to users. These classes help organize the system's data and responsibilities, providing a conceptual foundation for database design and application development.

Table 3.1 Core conceptual classes

Class Name	Description
User	Manages login credentials and user roles.
Student	Stores student information and submitted OD requests.
Faculty	Reviews OD requests and verifies certificates.
HOD	Handles OD approvals and certificate verification according to assigned permissions.
ODRequest	Stores OD request details and approval status.
Document	Represents uploaded OD letters and certificates.
CertificateValidation	Maintains certificate verification details and status.
Notification	Represents email notifications for request updates.

CHAPTER 4

METHODOLOGY AND PROJECT DESCRIPTION

4.1 General

The development of the OD Tracking Application followed a structured sequence, with each stage building on the previous one. Requirement analysis was completed first to identify the system's needs, followed by system design to define the architecture and workflow. Technology selection was then carried out based on the requirements of each component. Module implementation, integration, and testing followed to ensure that the application functions as intended. The objective of this phase is to establish a functional and reliable foundation for OD request management, approval, and certificate verification.

4.2 Requirement Analysis

Requirement analysis was carried out to identify the functional and technical needs of the OD Tracking Application. The process focused on understanding the difficulties in manual OD request handling, approval delays, document management, and certificate verification. Based on these needs, the system requirements were defined for student request submission, faculty and HOD approvals, status tracking, certificate uploads, and email notifications. These requirements formed the basis for system design and module development, ensuring that the application addresses the key challenges of the existing process.

4.3 Functional Requirements

The functional requirements define the essential operations that the OD Tracking Application must perform. The system allows students to log in, submit OD requests with supporting documents, track request status, and upload participation certificates. Faculty members and HODs can review requests, approve or reject applications, and verify submitted certificates according to their assigned roles. The

application also maintains relevant records and sends email notifications about approval decisions. These requirements ensure that the complete OD management process is handled through a structured and centralized platform.

Table 4.1 Functional requirements

ID	Requirement
FR1	The system shall allow students to log in using their credentials.
FR2	The system shall maintain student information and account details.
FR3	The system shall allow students to submit OD requests with event details and supporting documents.
FR4	The system shall allow faculty members to review submitted OD requests.
FR5	The system shall allow authorized faculty members to approve or reject OD requests.
FR6	The system shall allow the HOD to review and approve OD requests according to the approval workflow.
FR7	The system shall display the current status of each OD request to students.
FR8	The system shall allow students to upload participation certificates after completing the event.
FR9	The system shall allow faculty members or HODs to verify uploaded certificates.
FR10	The system shall send email notifications regarding OD request approval or rejection.
FR11	The system shall store OD requests and related documents for future reference.
FR12	The system shall allow authorized users to access and manage records according to their roles.
FR13	The system shall maintain the OD request and certificate verification process through a centralized application.

4.4 Non-Functional Requirements

The non-functional requirements define the quality attributes and performance expectations of the OD Tracking Application. The system should provide a user-friendly interface, reliable request processing, secure access to student and faculty information, and efficient retrieval of OD records. It should maintain data consistency during request approvals and certificate verification, support smooth interaction between the frontend and backend, and provide maintainability for future enhancements. These requirements help ensure that the application remains usable, secure, and dependable for students, faculty members, and HODs.

Table 4.2 Non-functional requirements

Requirement	Description
Performance	Efficient processing of OD requests, approval updates, and certificate verification.
Security	Protection of user accounts and application data through authentication and role-based access control.
Accessibility	Availability of the application to students, faculty members, and HODs through a web browser.
Reliability	Consistent handling of OD requests, approval decisions, and certificate records.
Scalability	Ability to accommodate additional students, faculty members, and OD requests as usage grows.
Privacy	Protection of student details, OD letters, and uploaded certificates from unauthorized access.
Usability	A simple and user-friendly interface that allows users to submit requests and track their status easily.
Maintainability	Application structure that supports bug fixes, updates, and future enhancements.
Portability	Access to the application through commonly used web browsers on supported devices.

4.5 Technology Selection

The technologies selected for the OD Tracking Application support the development of a web-based system for OD request submission, approval management, and certificate verification. HTML5, CSS3, and JavaScript are used to develop the frontend interface, providing structure, styling, and interactive features. Python with the Flask framework is used for backend development to handle application logic, user requests, approval workflows, and document processing. Azure SQL is used to store user information, OD request details, approval statuses, and related records. Email integration supports notifications regarding approval decisions and request updates. These technologies work together to provide a centralized platform that simplifies OD management and supports future improvements.

Table 4.3 Technology stack

Component	Technology	Role
Frontend development	HTML5	Defines the structure and content of the application's web pages.
User interface styling	CSS3	Provides visual styling, layouts, and a consistent interface.
Client-side interaction	JavaScript	Handles interactive elements and dynamic user interface updates.
Backend development	Python	Implements application logic and processes user requests.
Application framework	Flask (Python)	Connects frontend requests with backend operations and manages application routes.
Database management	Azure SQL	Stores student details, faculty information, OD requests, and approval records.
User authentication	Python and Flask	Handles login processing and supports access control based on user roles.
Document management	File upload and storage mechanism	Handles OD letters and participation certificates submitted by students.
Email notification	Email integration	Sends notifications about OD request approvals, rejections, and relevant status updates.
Web access	Web browser	Allows students, faculty members, and HODs to access the application.

The rationale for selecting the major technologies and models used in the project is presented in Table 4.4.

Table 4.4 Technology selection rationale

Technology	Reason for selection
HTML5	Provides a standard structure for web pages and supports the development of accessible application interfaces.
CSS3	Enables consistent styling, organized layouts, and an attractive user interface.
JavaScript	Supports interactive elements and dynamic updates that improve the user experience.
Python	Provides a simple and flexible programming language for implementing application logic and processing requests.
Flask	Offers a lightweight web framework for handling routes, processing requests, and connecting frontend operations with backend functionality.
Azure SQL	Provides flexible document-based storage for student details, OD requests, approval records, and certificate information.
User Authentication	Helps verify user identities and supports controlled access to application features.
File Upload and Storage	Enables students to submit OD letters and participation certificates electronically for review and verification.
Email Integration	Enables automatic communication of OD request decisions and status updates to users.
Web Browser	Allows students, faculty members, and HODs to access the application without installing separate desktop software.

4.6 System Modules

The implementation of the OD Tracking Application is divided into separate modules, each responsible for a specific stage of the OD management process. This modular structure simplifies development, testing, maintenance, and integration. Each module handles a defined set of operations, such as student management, OD request submission, faculty and HOD approvals, certificate verification, and email notifications. The modules work together through the application backend and database to provide a centralized workflow. This approach also allows individual modules to be modified or enhanced without significantly affecting the other components of the system. Table 4.5 presents the system modules and their responsibilities.

Table 4.5 System module descriptions

Module	Description
Student Management Module	Maintains student account details and relevant information required for OD request submission and tracking.
OD Request Submission and Tracking Module	Allows students to submit OD requests with supporting documents and monitor their request status.
Faculty Approval and Review Module	Enables faculty members to review OD requests and approve or reject them according to their responsibilities.
HOD Management and Approval Module	Supports HOD review and approval of OD requests within the defined approval workflow.
Certificate Upload and Verification Module	Allows students to upload event participation certificates and enables authorized staff to verify the submitted documents.
Email Notification Module	Sends email notifications to communicate OD request decisions and relevant status updates.

4.6.1 Student Management Module

The Student Management Module maintains student account details and the information required for OD request processing. It allows authorized administrators or faculty members to manage student records and associate each student with the appropriate account. The module helps ensure that OD requests are linked to the correct student details, reducing errors during request submission and approval. By maintaining organized student information, this module supports efficient record management and simplifies the retrieval of student-related OD details.

4.6.2 OD Request Submission and Tracking Module

The OD Request Submission and Tracking Module enables students to submit on-duty requests through the web application. Students provide the necessary event details and upload supporting OD letters or documents for review. Once submitted, the request is recorded in the system and made available to the authorized faculty members for processing. The module allows students to monitor the progress of their requests through status updates, helping them identify whether an application is pending, approved, or rejected without repeated manual follow-ups.

4.6.3 Faculty Approval and Review Module

The Faculty Approval and Review Module handles the initial review of OD requests submitted by students. Faculty members can access the relevant request details and supporting documents to evaluate the purpose and validity of each application. Based on the information provided, they can approve or reject requests according to their assigned responsibilities. The module updates the request status and supports communication of the decision through the application's notification process. This structured approach improves the organization of approvals and reduces dependence on paper-based procedures.

4.6.4 HOD Management and Approval Module

The HOD Management and Approval Module supports the review and approval responsibilities of the Head of the Department. It provides access to OD request information required for departmental review and allows the HOD to make approval decisions according to the configured workflow. The module maintains the relevant decision and status information so that request progress can be tracked consistently. By incorporating HOD review into the centralized application, the system helps organize departmental approval activities and improves the accessibility of OD request records.

4.6.5 Certificate Upload and Verification Module

The Certificate Upload and Verification Module manages the submission and validation of participation certificates after students complete their events. Students can upload the required certificate or proof of participation through the application. The uploaded document is associated with the relevant OD request so that authorized faculty members or the HOD can review it. After examining the submitted proof, the authorized reviewer can update its verification status. This module centralizes certificate handling, reduces dependence on physical documents, and makes verification records easier to access when required.

4.6.6 Email Notification Module

The Email Notification Module communicates important OD request updates to users through email. It supports notifications regarding approval or rejection decisions and other relevant changes in request status, depending on the application's implementation. By providing updates through email, the module helps students stay informed about their applications without needing to repeatedly contact faculty members or check manually. Integration with the backend allows notifications to be triggered by relevant workflow events, improving communication between students and the staff responsible for OD request processing.

4.7 OD Request Workflow and Status Mapping

The OD request workflow describes the process from application submission to certificate verification. Students log in, submit OD requests with supporting documents, and track their status. Faculty members and, where required, the HOD review requests and approve or reject them. The system updates the request status and sends relevant email notifications. After the event, students upload participation certificates for verification by authorized staff. The final verification status is recorded in the system, ensuring organized and centralized OD management.

Table 4.6 OD Request Status Mapping

Stage	Recognized input/status	System action
OD Request	Request submitted by student	Record the OD request
Faculty Review	Request awaiting faculty review	Display request for review
Approval	Request approved	Update status to Approved
Rejection	Request rejected	Update status to Rejected
Status Tracking	Current request status	Display status on student dashboard
Certificate Upload	Participation certificate submitted	Store certificate with the OD request
Certificate Verification	Certificate awaiting verification	Allow authorized staff to review and update verification status

Stage	Recognized input/status	System action
Email Notification	Approval or rejection decision	Send email notification to the student

4.8 Module Integration

The modules of the OD Tracking Application are integrated through the Flask backend and Azure SQL database to support a complete OD management workflow. The student management module provides the required user information, while the OD request module handles application submission and status tracking. Faculty and HOD modules manage request reviews and approval decisions, and the certificate module supports document upload and verification. The email notification module communicates important status updates to students. This integration enables the modules to share relevant information and operate as a centralized system for managing OD requests and certificates.

4.9 System Configuration

Development and testing were carried out on an ordinary desktop machine. Tables 4.7 and 4.8 give the hardware and the software used, respectively.

Table 4.7 Hardware configuration

Component	Specification
Processor	Intel Core i5 or i7, or an equivalent AMD processor
RAM	6 GB and above
Hard disk	500 GB
Pointing device	Two or three button mouse

Table 4.8 Software configuration

Component	Specification
Operating system	Windows 10 or Windows 11
Programming language	Python
Framework	Flask (Python)
Development environment	Python version 3.10.8

4.10 Feasibility Study

Feasibility was assessed before development across three dimensions: economic feasibility, technical feasibility, and operational feasibility. These evaluate the project's affordability, the availability of resources and technologies required for development, and the likelihood of successful adoption by its intended users.

4.10.1 Economical Feasibility

The OD Tracking Application is economically feasible because it uses open-source technologies such as Python, Flask, HTML, CSS, JavaScript, and Azure SQL. These technologies help reduce software development and licensing costs. The application can be developed and accessed using existing computers and web browsers, minimizing the need for additional hardware. Therefore, the system provides a cost-effective solution for managing OD requests, approvals, and certificate verification.

4.10.2 Technical Feasibility

The OD Tracking Application is technically feasible because it uses widely available technologies such as HTML, CSS, JavaScript, Python, Flask, and Azure SQL. It can be developed and accessed using standard computers and web browsers without specialized hardware. The selected technologies support request processing, database management, document uploads, and email notifications, making them suitable for implementing the system's core functions.

4.10.3 Social Feasibility

The social feasibility of the OD Tracking Application depends on its ease of use and acceptance by students, faculty members, and HODs. The application provides a simple web interface that allows users to submit OD requests, review applications, track statuses, and verify certificates with minimal effort. Email notifications and centralized records improve communication and reduce the need for repeated

manual follow-ups. These features make the system suitable for supporting the existing OD management process in an educational institution.

4.11 Testing Methodology

Testing is performed to ensure that the OD Tracking Application functions correctly and meets its requirements. The testing process examines individual modules, interactions between components, and the overall application workflow. It verifies OD request submission, approval and rejection, status tracking, certificate uploads, certificate verification, and email notifications. The results help identify errors and ensure that the system operates as expected.

4.11.1 Unit Testing

Unit testing focuses on verifying the individual modules of the OD Tracking Application independently. Each module, including student management, OD request submission, approval processing, certificate uploads, and email notifications, is checked to ensure that its functions work as intended. This testing helps identify errors within individual components before they are integrated into the complete system.

4.11.2 Integration Testing

Integration testing verifies whether the different modules of the OD Tracking Application work together correctly. It checks the interaction between the frontend, Flask backend, Azure SQL database, and email notification service. The testing focuses on ensuring that OD request details are stored correctly, approval decisions update request statuses, uploaded certificates are associated with the appropriate requests, and relevant notifications are triggered. This helps identify communication errors between integrated components.

4.11.3 Functional Testing

Functional testing verifies whether the OD Tracking Application performs its intended operations according to the specified requirements. It checks user login, OD request submission, faculty and HOD approval or rejection, request status

tracking, certificate uploads, certificate verification, and email notifications. Each function is evaluated using relevant input and expected outcomes to identify errors and confirm that the application behaves as intended.

4.11.4 System Testing

System testing evaluates the OD Tracking Application as a complete system to ensure that all modules function together correctly. It verifies the overall workflow, including user login, OD request submission, approval decisions, status tracking, certificate verification, and email notifications. This testing helps identify issues affecting the application's overall functionality and confirms that the system meets its intended requirements.

4.11.5 White Box and Black Box Testing

White box testing examines the internal logic and code structure of the OD Tracking Application, including request processing, approval workflows, authentication, and database operations. Black box testing evaluates the application's functionality by providing inputs and checking the resulting outputs without examining the internal code. Both approaches help identify errors, verify expected behavior, and improve the reliability of the application.

Table 4.9 Test cases

Test	Input	Expected output	Result
User login	Valid login credentials	User successfully accesses the dashboard	Pass
OD request submission	Event details and supporting OD letter	OD request is submitted and stored	Pass
Faculty review	Submitted OD request	Request details are displayed for review	Pass
Approval and rejection	Faculty/HOD decision	Request status is updated accordingly	Pass
Request status tracking	Submitted OD request	Current request status is displayed to the student	Pass
Certificate upload	Participation certificate file	Certificate is uploaded and associated with the request	Pass

Test	Input	Expected output	Result
Certificate verification	Uploaded participation certificate	Authorized reviewer can verify the certificate	Pass
Email notification	Approval or rejection decision	Relevant email notification is triggered	Pass

4.12 Initial Results

The initial development of the OD Tracking Application establishes a centralized platform for OD request submission, approval management, and certificate verification. The application is designed to support student login, request status tracking, faculty and HOD reviews, document uploads, and email notifications. These features aim to reduce manual paperwork, improve communication, and simplify the management of OD records. Further testing is required to evaluate the system's reliability and confirm the correct functioning of all implemented features.

4.13 Challenges

During the development of the OD Tracking Application, key challenges included implementing role-based access control, maintaining accurate request statuses, managing uploaded documents, and coordinating faculty and HOD approval workflows. Integrating email notifications and ensuring that certificates were associated with the correct OD requests also required careful implementation. Addressing these challenges helped establish a structured approach to OD request management, document handling, and communication within the application.

CHAPTER 5

CONCLUSIONS AND FUTURE ENHANCEMENTS FOR PHASE II

5.1 Conclusion

The OD Tracking Application provides a centralized web-based solution for managing on-duty requests, approval workflows, and certificate verification. It aims to reduce paperwork, simplify communication between students and faculty, and improve access to OD records. By integrating request submission, approval management, status tracking, document uploads, and email notifications, the system provides a structured approach to OD management. The application also provides a foundation for future improvements and additional functionalities.

5.2 System Implementation Status

1. Problem definition and system requirements established for digital OD request management.

2. Literature survey completed, covering web-based student information systems, document management, and workflow automation.

3. System architecture established with frontend, backend, database, and email notification components.

4. HTML5, CSS3, and JavaScript selected for frontend development.

5. Python and Flask selected for backend development and request processing.

6. Azure SQL is selected for storing student information, OD requests, approval records, and related data.

7. User authentication and role-based access established for students, faculty members, and HODs.

8. OD request submission, faculty review, approval, rejection, and status tracking functionalities developed.

9. Certificate upload and verification functionalities incorporated into the OD management workflow.

10. Functional and non-functional requirements documented, with testing planned to evaluate the application's correctness and reliability.

5.3 Limitations

The current system has certain limitations that may be addressed through future development. Access depends on the availability of the deployed application and a compatible web browser. Certificate verification requires manual review by authorized faculty members or the HOD. Additional reporting, analytics, automated reminders, and advanced document validation may also be needed to improve the system's capabilities.

5.4 Future Development Objectives

Future development aims to improve the functionality, usability, security, and efficiency of the OD Tracking Application. Key objectives include refining the approval workflow, enhancing document management, introducing automated reminders, and improving reporting facilities. Further improvements may also focus on system performance, user experience, and the management of larger volumes of OD requests.

5.5 Future Enhancements

Future enhancements may include developing a mobile application, introducing automated reminders for pending requests, and providing dashboard analytics for OD request statistics. Advanced certificate validation and downloadable reports could improve document verification and record management. Integration with attendance management systems may also help connect approved OD requests with student attendance records.

5.6 Future Development Schedule

Table 5.1 Future enhancements

S.No	Planned activity	Expected outcome
1	Requirement Review	Identify areas for improvement in the existing application.
2	System Enhancement	Refine the user interface and approval workflow.
3	Feature Development	Introduce selected features such as reminders and reporting.
4	Module Integration	Integrate additional features with existing modules.
5	Testing and Validation	Evaluate functionality, security, and reliability.
6	Documentation	Update the technical documentation and project report.

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CO–SDG RELEVANCE RECORD

Project Title : ON DUTY TRACKING SYSTEM

Batch Members : SHARJIN JINO S A (241701052)

NAVEEN S (241701038)

AJAY RAM IYYAPPAN (241701004)

Supervisor : Dr. P. REVATHY, Ph.D.

SDG (No. & Theme)	Addressed COs	Topic/Activity addressing SDG Theme
SDG 4 – Quality Education	CO3, CO5	Supports educational administration by simplifying OD request submission, approval, and certificate verification through a centralized digital platform.
SDG 9 – Industry, Innovation and Infrastructure	CO3, CO5	Applies web technologies and database management to develop an innovative digital solution for automating institutional OD management.
SDG 10 – Reduced Inequalities	CO2, CO4	Provides a consistent web-based process for students to submit requests and track approval status, improving access to OD management services.
SDG 16 – Peace, Justice and Strong Institutions	CO1, CO5	Promotes transparent and accountable institutional processes through organized approval workflows, status tracking, and centralized record management.

Signature of the Supervisor

COURSE OUTCOMES

CO1: Construct design challenge and reframe the design challenge into design opportunity.

CO2: Interview the user, and know the feelings of users to foster deep user understanding and be able to uncover the deep user insights and needs.

CO3: Develop ideas and prototypes by brainstorming

CO4: Organize the user walkthrough experience to test prototype

CO5: Develop smart strategies and implementation plan that will deliver/achieve the idea/solution deduced from earlier phases

PROGRAM OUTCOMES

PO	Outcome	Description
PO1	Engineering Knowledge	Apply knowledge of mathematics, science, engineering fundamentals and an engineering specialization to the solution of complex engineering problems.
PO2	Problem Analysis	Identify, formulate, review research literature and analyse complex engineering problems to reach substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences.
PO3	Design / Development of Solutions	Design solutions for complex engineering problems and design system components or processes that meet specified needs with appropriate consideration of public health and safety, and cultural, societal and environmental considerations.
PO4	Conduct Investigations of Complex Problems	Use research-based knowledge and research methods, including design of experiments, analysis and interpretation of data and synthesis of information, to provide valid conclusions.
PO5	Modern Tool Usage	Create, select and apply appropriate techniques, resources and modern engineering and IT tools, including prediction and modelling, to complex engineering activities with an understanding of the limitations.
PO6	The Engineer and Society	Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to professional engineering practice.
PO7	Environment and Sustainability	Understand the impact of professional engineering solutions in societal and environmental contexts and demonstrate the knowledge of, and need for, sustainable development.
PO8	Ethics	Apply ethical principles and commit to professional ethics and responsibilities and norms of engineering practice.
PO9	Individual and Team Work	Function effectively as an individual, and as a member or leader in diverse teams and in multidisciplinary settings.
PO10	Communication	Communicate effectively on complex engineering activities with the engineering community and with society at large, including writing effective reports and design documentation and giving clear instructions.
PO11	Project Management and Finance	Demonstrate knowledge and understanding of engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects in multidisciplinary environments.
PO12	Life-long Learning	Recognize the need for, and have the preparation and ability to engage in, independent and life-long learning in the broadest context of technological change.

PROGRAM SPECIFIC OUTCOMES

PSO	Area	Description
PSO1	Software Development and Database Management	Strength in the foundations of programming languages and competence in computing technologies and tools to design and implement efficient software solutions using suitable algorithms, data structures and other computing techniques
PSO2	Human-Computer Interaction (HCI) and Web Application Design	A skill to Independently investigate problems which can be solved by a Human Computer Interaction (HCI) design process and design an end-to-end solution from user need identification to UI design to technical coding and evaluation. Ability to effectively use suitable tools and platforms, as well as enhance them, to design and develop applications/products in animation, gaming, augmented and virtual reality, etc.
PSO3	Innovation and Digital Transformation in Education	An ability to apply knowledge in various domains to identify research gaps and to provide solutions to new ideas, inculcate passion towards higher studies, creating innovative career paths to be an entrepreneur and evolve as an ethically social responsible computer science and design professional.

CO–PO–PSO MAPPING

Project Title : ON DUTY TRACKING SYSTEM

Batch Members : SHARJIN JINO S A (241701052)

NAVEEN S (241701038)

AJAY RAM IYYAPPAN (241701004)

Supervisor : Dr. P. REVATHY, Ph.D.

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	2	3	3	2	2	2	1	2	--	2	--	2	2	2	2
CO2	2	3	2	2	2	2	1	2	2	2	--	2	1	2	2
CO3	3	2	3	2	3	2	1	2	3	2	2	2	3	3	2
CO4	2	2	2	3	2	2	1	2	3	3	–	2	2	3	1
CO5	2	2	3	2	3	2	2	3	2	2	3	3	3	3	3

1 – Slight (Low); 2 – Moderate (Medium); 3 – Substantial (High). If there is no correlation, put “--”.

Signature of the Supervisor